Hidden Deadly Factors in Everyday Life: A review

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The work presents the authors' systematization of everyday environment factors with a non-zero probability of a delayed lethal effect. Several groups are distinguished: harmful organic compounds, harmful external influences, and harmful additives. The review is aimed at the actual detection of these factors under domestic conditions: where, in what form, and in what quantity their presence has been recorded in the literature. For organic compounds, the pathways and numerical data on the formation of volatile organic compounds, aldehydes, acrylamide, and polycyclic aromatic hydrocarbons are considered during heating (Maillard reaction, thermal oxidation, pyrolysis), under the action of household oxidizing agents (sodium hypochlorite, ozone), and at room temperature. Data are presented on VOC emission from paints, adhesives, varnishes, metalworking fluids, 3D printing materials, and particleboard. Data are presented for external influences such as UV and ionizing radiation. Literature data are shown for structural additives such as nickel, vanadium, cobalt, fiberglass, and asbestos. For hexavalent chromium, the conditions of its formation at high temperatures are given. For the remaining additives, the recorded concentrations and detection frequencies are provided: nitrosamines in sausages, polychlorinated biphenyls in paint pigments, bisphenol A in canned food, phthalates in ready-to-eat meals, and brominated flame retardants in household electronics, toys, and building materials.

This work is a preprint version, it will be supplemented further.

Keywords: volatile organic compounds, aldehydes, acrylamide, polycyclic aromatic hydrocarbons, ozone, ultraviolet radiation, ionizing radiation, radon, hexavalent chromium, hazards, toxicity, domestic environment, 3d printing.

Introduction

The everyday human environment consists of natural factors and products of technological progress. However, among them there are also those capable of causing harm to health, including factors with a non-zero probability of a delayed lethal effect. The toxicity of many of them belongs to fundamental and widely known scientific knowledge, which emphasizes the critical importance of systematizing the pathways by which they enter human everyday life. Such factors can be conditionally divided into several groups: harmful organic compounds, harmful external influences, and harmful

additives. This division is not an official classification; however, it is adopted by the authors for convenience of systematization.

The subject of this review is the actual detection of toxic factors under conditions of the human everyday environment. Sensitivity to them is individual, and permissible levels differ between jurisdictions and are revised as data accumulate. Therefore, the authors deliberately shift the emphasis from the question of what quantity is permissible to the question of where and in what form the presence of these factors has been recorded. Establishing threshold values permissible for humans is outside the scope of this review. Interpretation of the findings in terms of safety remains the domain of specialists in the relevant field.

Harmful Organic Compounds

Organic compounds are defined as compounds of carbon, typically in combination with hydrogen, oxygen, and nitrogen. Organic objects in everyday human life are so numerous that listing them would be impractical; it’s more convenient to go the other way and name what isn’t organic. Inorganic substances include simple substances (metals and nonmetals, including graphite, diamond, oxygen, and nitrogen in air), metal alloys, inorganic compounds (water, oxides, salts, minerals, and the simplest carbon compounds such as CO_2 and carbonates), and inorganic materials (glass, ceramics, concrete). The list is incomplete, but it covers what’s relatively frequent in everyday life. Everything else that surrounds a person is, with high probability, organic. Including materials that may not outwardly look the part. Composite products that visually resemble metal, for instance, may actually be polymer-based [1]. Kardos et al. show this directly: commercial PLA-based composites filled with copper and bronze powders impart a metallic appearance and texture, while the matrix itself stays polymeric, and the metal particles (3–5 μm for the copper filler, 6–30 μm for the bronze filler) are simply distributed within it, with weak adhesion to the matrix in a number of these materials.

Harmful organic compounds formed under everyday conditions include primarily volatile organic compounds (VOCs), including aldehydes, as well as acrylamide and polycyclic aromatic hydrocarbons (PAHs). Many of them arise in everyday life during the heating of organic materials and food products, which promotes several processes: the Maillard reaction [2], oxidation [3], and pyrolysis [4].

The Maillard reaction is a non-enzymatic interaction of amino groups of amino acids, peptides, and proteins with carbonyl groups of reducing sugars [2]. In food products simultaneously containing proteins or amino acids and reducing sugars (glucose,

fructose, lactose, etc.), it proceeds most intensively during thermal processing, but is also possible during prolonged storage. A typical visual manifestation of the reaction is the formation of a golden-brown crust [5]. Among the products of this reaction, toxic aldehydes have been identified [6]. Cha et al. found that heating (160 °C) systems of “reducing sugar – amino acid” produces glyoxal (5.92–39.1 μg/mL) and methylglyoxal (3.66–151.88 μg/mL), with higher content in monosaccharide systems than disaccharide ones; the maximum methylglyoxal (151.88 μg/mL) came from the fructose–threonine system. Acrylamide can also result from this reaction [7]. Here asparagine is the key player, per Mottram et al. [7] — heating asparagine alone, without sugar, produces no acrylamide. According to [8], acrylamide content in Spanish-market potato chips (80 samples, 18 manufacturers) ranged from 108 to 2180 μg/kg, averaging 630 μg/kg with a median of 556 μg/kg.

Oxidation is a reaction in which a substance donates electrons to an oxidizing agent — most often, in everyday life, atmospheric oxygen. Frying is one example: the thermal oxidation of fats and oils. According to [9], this produces hydroperoxides, which then break down into low-molecular-weight volatile compounds (aldehydes, ketones, carboxylic acids, short-chain alkanes and alkenes), as well as dimers and polymers of triacylglycerols formed via radical reactions and Diels–Alder reactions.

During oven roasting of beef (250 °C, 3×3×2 cm pieces), hexanal rose from 122.67 μg/kg in raw meat to a peak of 811.42 μg/kg after 6 minutes, then fell to 196.56 μg/kg by minute 18 [10]. Nonanal rose from 44.84 to 179.55 μg/kg, octanal from 8.64 to 70.46 μg/kg, and heptanal from 0 to 69.69 μg/kg. Hexanal, heptanal, octanal, and nonanal were the dominant aldehydes. Toluene, meanwhile, kept climbing: 9.32 to 446.34 μg/kg by the 18th minute.

Pyrolysis — chemical reactions in organic substances at high temperature without oxygen, or with too little of it — leads to new and often more complex compounds. This is what happens when food is cooked over coals or open fire, and one result can be PAHs. According to [11], total PAH content (16 compounds) in grill-cooked steaks, across different charcoal types, ranged from 23.80 to 36.39 μg/kg.

Oxidation doesn’t require oxygen or heat, though. In everyday life the oxidizing agent might instead be something like sodium hypochlorite [12]. Household bleach based on NaOCl (pH around 12) releases hypochlorous acid (HOCl) and chlorine (Cl_2) into the gas phase during use — both strong oxidizers.

Ozone, too, can act as an oxidizer. It's an allotropic form of oxygen (O_3), and its reactivity comes from the molecule's thermodynamic instability — an oxygen atom detaches easily, forming stable O_2. Waring and Siegel [13] recorded rising formaldehyde and nonanal concentrations (in some cases within meas-urement error) in a 27 m³ residential room while an ozone-releasing ion generator was running.

In addition to heating, a cooling system can also serve as a source of VOCs. Organic components of cooling agents, when in contact with a heated object, may undergo oxidation, degradation, and other thermally initiated transformations; the resulting VOC composition depends on the composition of the agent itself. Levilly et al. identified 15 carbonyl compounds among the VOCs generated during the use of metalworking fluids (MWFs). These included formaldehyde, acetaldehyde, acetone, acrolein, pro-panal, crotonaldehyde, butyraldehyde, benzaldehyde, isovaleraldehyde, valeraldehyde, o-tolualdehyde, m+p-tolualdehyde, glutaraldehyde, hexanal, and 2,5-dimethylbenzaldehyde [14]. In petroleum-based MWFs the median total aldehyde concentration was 32.4 µg/m³ (IQR 28.7–65.5). Formaldehyde ac-counted for 5.4 µg/m³ of that figure (IQR 3.9–7.0). Water-based MWFs had a higher median of 43.5 µg/m³ overall (IQR 24.7–77.2), with formaldehyde at 6.1 µg/m³ (IQR 3.5–8.0).

Room-temperature evaporation is another, independent release pathway for VOCs. Stockwell et al. measured a solvent-borne polyurethane paint (primer) releasing 495 g of VOCs per kg of product upon evaporation, accounting for 83% of the total mass lost [15]. Other coatings examined in the same study emitted substantially less. Latex paint released 43.1 g/kg (6.63%), primer/sealant 2.84 g/kg (0.66%), and "zero VOC" paint only 0.71 g/kg (0.17%). The ranking of peak emission rates matched: 6.69 mg/h for polyurethane paint, 3.16 mg/h for latex, 0.14 mg/h for "zero VOC" paint, and 0.03 mg/h for pri-mer/sealant. More than 30 hydrocarbon peaks were detected in the solvent-borne paint's emissions, dom-inated by hydrocarbons and aromatic compounds such as toluene, C8 aromatics, and para-chlorobenzotrifluoride. Water-based paints emitted mostly small oxygen-containing VOCs — methanol, ethanol, acetone, formaldehyde, ethylene glycol — plus Texanol, which made up 13% of the pri-mer/sealant's output. In both cases, more than half of the total VOC mass had escaped within the first one to two hours.

Among 25 adhesives tested by Kozicki and Guzik in chamber conditions (23 ± 1 °C, 50 ± 5% relative humidity, 0.1 m³ chamber, air exchange rate 0.5 h^{-1}, ISO 16000-9), solvent-borne types produced the highest emissions [16]. Average TVOC reached 8451 µg/m³ after 3 days and fell to 2214 µg/m³ by day 28. Floor adhesives in this category

ranged from 675 to 4184 μg/m³. Finishing adhesives ranged lower, from 673 to 1490 μg/m³. After 3 days, one synthetic-rubber floor adhesive in organic solvents con-tained n-hexane at 816 μg/m³, methylcyclopentane at 823 μg/m³, 3-methylhexane at 720 μg/m³, n-heptane at 446 μg/m³, and 2,3-dimethylbutane at 239 μg/m³. A synthetic-resin adhesive in organic sol-vents was different: it consisted mainly of acetone, at 458 μg/m³. Water-dispersion adhesives released far less formaldehyde — 10 μg/m³ for finishing adhesive, 9–19 μg/m³ for wall and ceiling adhesives at 3 days — and this fell below 1 μg/m³ by day 28.

Stachowiak-Wencek and Prądzyński turned to water-based acrylic varnishes applied to oak, documenting a markedly different VOC profile [17]. By 24 hours, 2-butoxyethanol in varnishes A and B had reached 231.7–1531.0 μg/m³. Other compounds measured lower: n-decane at 60.2 and 149.4 μg/m³, n-nonane at 10.0 and 37.8 μg/m³, toluene at 3.8–5.0 μg/m³, o-xylene at 13.1 μg/m³, m-xylene at 5.3 μg/m³, and 1,3,5-trimethylbenzene at 7.3 μg/m³, with 1,2,3-trimethylbenzene and 1-ethyl-4-methylbenzene adding 26.4 μg/m³ together. TVOC reached 388–1794 μg/m³. The same first author, this time with Mateńko-Nożewnik, extended this work to UV-curable water-based varnishes [18]. Separate samples, measured at the same 24-hour mark, contained benzaldehyde (98.5–152.4 μg/m³), cyclohexanone (10.0–89.2 μg/m³), 1-butanol (30.9–61.2 μg/m³), 1,4-dioxane (8.2–31.0 μg/m³), toluene (9.6–12.8 μg/m³), and butyl acetate (11.5–12.5 μg/m³). TVOC here fell between 251 and 557 μg/m³. The same study also found hexanal (13.1–37.2 μg/m³) and furfural (4.7–11.2 μg/m³) in unvarnished wood, at 36.2–49.8 μg/m³ and 31.2–52.1 μg/m³ respectively. It is therefore unclear whether these two compounds originate from the varnish or the wood itself. In conversion varnishes, Howard et al. recorded formaldehyde emission 2.3–8.1 times above the free formaldehyde content of the applied product, with an emission rate of 0.17 mg/m²·h at 115 days [19]. McCrillis et al. measured free formaldehyde content in varnishes at 0.15–0.58 wt.%. The main volatile components they identified were methyl ethyl ketone, isobutanol, n-butanol, methyl isobutyl ketone, toluene, ethylbenzene, xylenes, and 1,2,4-trimethylbenzene [20].

3D printing presents its own emission profile. FDM printing releases VOCs mainly through polymer heating. SLA printing, by contrast, generates emissions tied to the components used in photopolymer resins. Zhang, Davis, and Black measured SLA’s total VOC emission rate at three to six times that of FFF. It exceeded 4 mg/h. The mixture itself was complex, made up of acrylates, aldehydes, alcohols, ketones, and hydrocarbons [21].

Formaldehyde in particleboard comes from the urea-formaldehyde (UF) resins used to make it. Que et al. studied these resins and found a molar ratio of formaldehyde to urea between 0.97 and 1.27 [22]. Using the perforator method, Kelleci et al. measured formaldehyde emission at 2.5–5.1 mg/100 g for particleboard based on foamed UF resins, rising to 3.95–7.31 mg/100 g for non-foamed resins [23].

Harmful External Influences

Such influences are ultraviolet and ionizing radiation. Short-wavelength UV radiation photolyzes molecular oxygen to form ozone, which makes it a double factor: through its own action and through the ozone formed. Stinson, Zhang, and Xia [24] showed that the photodissociation of O_2 at 185 nm yields two O (^3P) atoms, which produce ozone; for this, a 6 W low-pressure mercury lamp was used with an intensity at a distance of 20 cm of 42 μW/cm² at 185 nm and 73 μW/cm² at 254 nm. Szeto et al. [25] state that ozone-generating lamps of this type emit mainly at 254 nm, and the power of the 185 nm line is about 10% of that. At the same time, Martínez de Alba et al. [26] found that the peak ozone concentration on a surface at a distance of 20 cm from a robot with a 450 W quartz lamp (185–254 nm) was 0.37 ppm. And finally, Claus [27] notes that wavelengths shorter than 240 nm generate ozone more easily, while the 185 nm line produces ozone if it is not absorbed by the lamp envelope.

As regards UV sources, according to Yagura et al. [28], in the UV part of the solar spectrum at the Earth's surface, the share of UVA is ~95%, UVB ~5% (by radiation energy; UVC is completely absorbed by the atmosphere). Alhamdan, Spicer, and Cates [29] show that low-pressure mercury lamps emit mainly at the 254 nm line, the secondary 185 nm line provides about 10% of the photon output, and the ratio of these two lines depends on the lamp design (mercury vapor pressure, buffer gas, amalgam, quartz envelope transmission) and on the operating temperature. According to Ford et al. [30], manicure lamps emit ≈100% UVA. For solariums, according to Gerber et al. [31], these shares are ≈96–99.5% UVA, 0.5–4% UVB, and ≈0% UVC, respectively. Thus, according to the data, in modern tanning installations the share of UVB in the total UV radiation varies from 0.5 to 4%, while the rest is accounted for by UVA. For comparison, in old tanning lamps with a single mercury lamp (before the 1960s), the spectrum contained about 20% UVC and 30–50% UVB. Tanning products, meanwhile, may contain photoactive components that, under the action of UV radiation, generate strong oxidizers — reactive oxygen species and free radicals. For example, inorganic UV filters (TiO_2 and ZnO) are known to generate reactive oxygen species under the action of light, including short-lived free radicals, superoxide (O_2•–), and the hydroxyl radical (OH•)

[32]. Vejerano et al. [32] found that the sunscreen formulation that produced the strongest radical signal contained 18.6 wt.% ZnO and 6.6 wt.% TiO.

Ionizing radiation is also present in the domestic environment. With increasing altitude above sea level, the attenuation of cosmic radiation by the atmosphere decreases, and the dose rate increases. Safitri, Miska, and Yusibani [33] report that the cosmic radiation dose rate at sea level is 0.03 μSv, while at an altitude of 10 km it reaches 2.38 μSv, that is, approximately 80 times higher. By contrast, in basements, a significant source of exposure is the natural radioactive gas radon (^{222}Rn) and its decay products [34]. Ptiček Siročić et al. [34] found that in basements in Northern Croatia (15 measurement points), the arithmetic mean of the radon activity concentration was 70.9 ± 38.8 Bq/m³ (geometric mean 61.2 Bq/m³), whereas on the first and second floors it was 42.5 ± 30.8 Bq/m³ (geometric mean 32.8 Bq/m³).

Harmful Additives

Among them, structural additives should first be highlighted: nickel, vanadium, cobalt, fiberglass, asbestos, and also hexavalent chromium. Their content is relevant to take into account before beginning machining, welding, and other production operations that generate aerosols. Wu et al., studying friction-stir-processed 34CrNiMo6 (a high-strength quenched and tempered medium-carbon low-alloy steel), report the following composition: 1.5% Ni (wt.%), 1.5% Cr, 0.25% Mo, 0.34% C, 0.50% Mn, and 0.40% Si [35].

Ferrous metallurgy consumes around 85% of vanadium-containing products. The element goes into high-strength low-alloy, tool, and rail steels, as well as steels for construction and automobiles, while the remainder is directed into non-ferrous alloys with titanium, nickel, aluminum, and chromium (in par-ticular, Al-V alloys for titanium alloys in the aerospace industry), into catalysts, and into flow vanadium batteries. These figures come from Moskalyk and Alfantazi's review of vanadium processing [36], sup-plemented by Santos et al.'s assessment of vanadium's role in decarbonization [37].

Lithium-ion batteries are the primary consumer of cobalt. In 2024, batteries made up close to 76% of global cobalt consumption, with electric vehicles alone responsible for an estimated 43% of that share. Cobalt also appears in high-speed steels for cutting tools, such as M42 (8% Co), with this class of steels generally containing between 5% and 12% Co, and in WC-Co hard-alloy tools, where it functions as the binder phase — for instance, WC-7 wt.% Co. This picture draws on Picazo-Rodríguez et al.'s re-view of cobalt metallurgy and recycling [38], Wang et al.'s work on Co's influence on M42

microstruc-ture [39], and Silva et al.'s study of multilayer diamond coatings for cemented-carbide milling [40].

Fiberglass, used mainly as a reinforcing filler for polymer composites (GFRP), accounts for the largest share by market volume among fiber-reinforced polymers, owing to the low cost of the fiber — a point made across the reviews by Silva et al. [40], Scaffaro, Di Bartolo, and Dintcheva on recycling of glass- and carbon-fiber composites [41], Mohammadi et al. on GFRP in automotive bumpers [42], and Sathishkumar, Satheeshkumar, and Naveen's broader review of glass-fiber-reinforced polymers [43]. In everyday life and closely related fields, such composites appear in electronics (printed circuit boards), the automotive industry (for example, bumpers made of polypropylene reinforced with 40% long glass fiber), construction, containers and tanks for chemical liquids, and sports equipment. In the form of glass wool it serves as thermal and acoustic insulation, including in buildings, as Ren et al. demonstrate in their work on glass-fiber-wool/MTMS aerogel composites with improved thermal insulation [44]. Kemari et al., reviewing high-temperature polymers for cable insulation, note that fiberglass can further raise the heat resistance of cable insulation [45]; and Levinson et al., studying airflow infiltration in ducts, report that a fiberglass lining for air ducts provides thermal and acoustic insulation, with an impermeable barrier over its air-facing surface helping to preserve that effectiveness [46].

Asbestos was historically widespread in construction as an effective insulator, and was also worked in-to fabrics, paper, cement, and plastics to impart structural strength, as detailed in Bolan et al.'s review of hazardous asbestos-containing materials [47]. About 80% of the asbestos imported by Japan in 2006 went into building materials (asbestos boards and cement products), about 7% into friction materials, and less than 3% into asbestos textile products; among the specific materials cited are asbestos-cement sheets and pipes, external roofing products made of asbestos-cement, and thermal insulation materials for buildings.

Chromium behaves differently depending on its oxidation state: stable in its trivalent form, it becomes a health hazard upon transition to the unstable hexavalent form. The transition Cr(III) → Cr(VI) is an ox-idation involving the loss of three electrons, a condition that can arise from machining metal without sufficient cooling. Van Leeuwen et al., reviewing the reactive evaporation and condensation of chromi-um, report that reactive evaporation of Cr from the surface oxide of stainless steels — forming volatile $CrO_2(OH)_2$ and CrO_3, i.e., Cr(VI) compounds — occurs above 500 °C in both humid and dry oxidizing environments [48]. Industrially, chromium is used predominantly as

an alloying element: about 90% of primary chromium goes into metallurgy, mostly into stainless and heat-resistant steels (roughly 86% of chromium directed to steel), while the rest is split between chemicals (about 6.5%) and refractories (about 3.5%), per Johnson, Schewel, and Graedel's account of the anthropogenic chromium cycle [49]. Mehtani et al. show that in iron-chromium alloys, oxidation slows sharply once Cr content reaches 12 wt.%, owing to the protective oxide film Cr_2O_3, while high-temperature iron alloys typically cap chro-mium at around 30% [50]. Guo et al. report 4.5–6.0% Cr in high-speed steels with boron used for hot-rolling rolls [51], and Chen et al. find 0–0.6% Cr in automotive dual-phase steel DP980, where it sup-presses bainitic transformation [52]. Among household items, Johnson et al. point to stainless steel kitchenware (about 18% Cr) [49].

A further group of additives worth noting includes nitrosamines, polychlorinated biphenyls (PCBs), bi-sphenol A, phthalates, brominated flame retardants, and heavy metals. Wang et al., analyzing commer-cial sausages on the Chinese market, found total N-nitrosamine content of 1.35–15.88 μg/kg (with one exception) and NDMA levels of 4.25–10.52 μg/kg in fermented sausages; in one sample (Shuanghui Corn Sausage), NPYR reached 71.73 μg/kg [53]. Hu and Hornbuckle, examining 33 commercial paint pigments, detected PCBs in 15 samples — predominantly organic pigments — in the range of 2–200 ng/g wet weight, identifying more than 50 congeners, with PCB11 present in 13 of the 15 PCB-containing samples [54]. Noonan, Ackerman, and Begley found bisphenol A in 71 of 78 canned food samples from U.S. retail, at concentrations of 2.6–730 ng/g [55]. Edwards et al., surveying 64 ready-to-eat meals from U.S. fast-food chains, detected di-n-butyl phthalate (DnBP) in 81% of samples and DEHP in 70%, with a median DEHP concentration of 36.0 μg/kg in hamburgers (method detection lim-it 14 μg/kg) [56]. Li et al., screening 120 consumer products with a portable X-ray fluorescence spec-trometer, found that about 37% contained at least one bromine-positive spot — most often in electrical appliances and toys — and went on to quantify PBDEs, HBCDD, and TBBPA in 15 of the bromine-positive samples [57]. Kajiwara, Noma, and Takigami report a similar pattern for new items on the Jap-anese market (electronics, curtains, wallpaper, building materials), which contained traditional bromin-ated flame retardants [58].

Conclusion

Thus, using the systematization proposed by the authors, the above-described factors encountered in everyday life can be divided into the following groups: harmful organic compounds, which will include predominantly VOCs, aldehydes, acrylamides, and

PAHs; harmful external influences, such as ultraviolet and ionizing radiation; and harmful additives. The factors that form harmful organic compounds include predominantly the processes of heating organic objects involving such reactions as the Maillard reaction, oxidation, and pyrolysis. The oxidizing agents are atmospheric oxygen, specialized chemical agents, and ozone. The sources of UV under domestic conditions are sunlight, germicidal lamps, manicure lamps, and solariums. The factor of ionizing radiation is the change in the altitude of the location, manifesting as an increase in cosmic radiation, or radon and its decay products. And harmful additives, in turn, are structural additives such as nickel, vanadium, cobalt, fiberglass, asbestos, and chromium in its hexavalent form, as well as other additives such as nitrosamines, heavy metals, polychlorinated biphenyls, bisphenol A, phthalates, and brominated flame retardants.

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